

Agenda

Hypothesis testing: short review with class exercise

Significance testing

Statistical hypotheses

A statistical hypothesis or hypothesis is a claim about the value of a single population characteristic or several population characteristics.

Definition: The null hypothesis denoted by H_0 , is the claim about one or more population characteristics that is initially assumed to be true. The alternative hypothesis, H_1 or H_a , is the assertion contradictory to H_0 .

When to formulate these hypotheses?

EXTERNAL EFFECTS (to the statistical process)

- A new product will be introduced, certain specifications needed, etc.
- Milemaster: introduce the tire or not?

CHANGE IN THE STATISTICAL PROCESS

- More complaints from your long-term customers on your product:
HAS THE DEFECT PROPORTION CHANGE?
- Are the fish safe? (HAS STH. CHANGED IN THE FISH?)

RESULT OF PHYSICAL/CHEMICAL MODELS

- the effect of increasing the amount of iron in a certain alloy on the tensile strength.

Analysis of hypotheses testing-I

The truth and falsity of a particular process can never be known with certainty!! (unless you examine the whole population)

Therefore: a hypotheses-testing procedure should be developed with the probability of reaching a wrong conclusion in mind!

Analysis of hypotheses testing - II

The null hypothesis H_0 : protected

H_0 will be rejected in favor of H_a only if sample evidence suggests H_0 is false.

If the sample does not strongly contradict H_0 , we continue to believe in the truth of H_0 .

Rejecting H_0 is a strong conclusion.

Accepting H_0 is a weak conclusion.

Hypothesis Testing

Based upon the sample evidence you decide to	The truth about the population mean	
	H_0 : True	H_0 : False
Do not reject H_0	Correct decision	Type II error
Reject H_0	Type I error	Correct decision

- Type-I error is more serious than type-II error.
- Hypotheses are chosen accordingly. $\rightarrow H_0$: protected

Computation of errors

$$\alpha = P\{\text{Type I}\} = P\{\text{reject } H_0 \text{ when } H_0 \text{ is true}\}$$

α : called significance level

$$\beta = P\{\text{Type II}\} = P\{\text{accept } H_0 \text{ when } H_0 \text{ is false}\}$$

→ depends on the exact parameter. “ H_0 is false” specifies a large parameter space.

1- β : power of the test

$$1- \beta = P\{\text{reject } H_0 \text{ when } H_0 \text{ is false}\}$$

General Procedure for Hypothesis Tests

1. From the problem context, identify the parameter of interest.
2. State the null hypothesis, H_0 .
3. Specify an appropriate alternative hypothesis, H_a .
4. Choose a significance level, α .
5. Determine an appropriate test statistic with the assumption that H_0 is true.
6. State the rejection region for the statistic.
7. Compute any necessary sample quantities, substitute these into the equation for the test statistic, and compute that value.
8. Decide whether or not H_0 should be rejected and report that in the problem context.

Significance levels reflect the seriousness of type-I error in a particular problem; the most common values are 0.1, 0.05, and 0.01.

Testing the population mean with a large sample

$$H_0: \mu = \mu_0$$

$$\text{Test statistics: } Z = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

Test statistics is given under the assumption that H_0 is true.

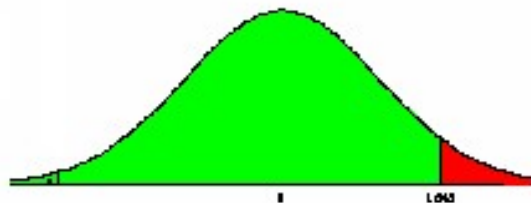
$$H_1: \mu > \mu_0$$

$$H_1: \mu < \mu_0$$

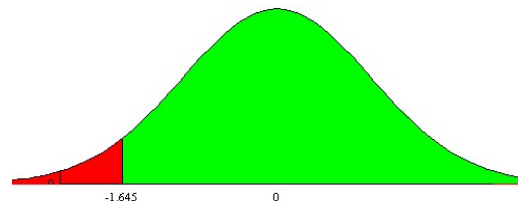
$$H_1: \mu \neq \mu_0$$

Accept

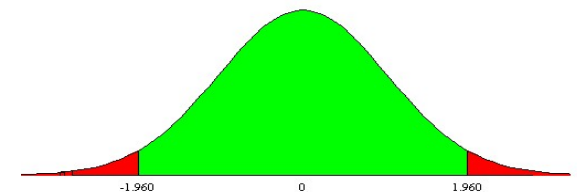
Reject



$$\text{R.R. : } z \geq z_\alpha$$



$$\text{R.R. : } z \leq -z_\alpha$$



$$\text{R.R. : } z \leq -z_{\alpha/2} \text{ or } z \geq z_{\alpha/2}$$

Tests about a population mean of a normal r.v. with known variance

$$H_0: \mu = \mu_0$$

$$\text{Test statistics: } Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

Test statistics is given under the assumption that H_0 is true.

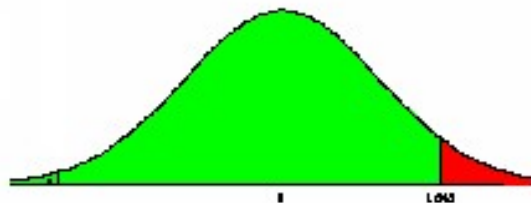
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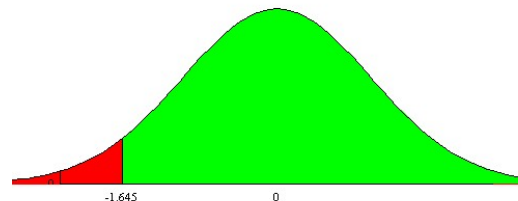
$$H_1: \mu \neq \mu_0$$

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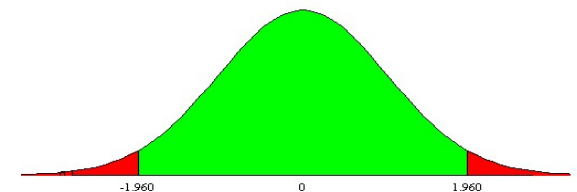
Reject



$$\text{R.R. : } z \geq z_\alpha$$



$$\text{R.R. : } z \leq -z_\alpha$$



$$\text{R.R. : } z \leq -z_{\alpha/2} \\ \text{or } z \geq z_{\alpha/2}$$

Class exercise:

Cigarette manufacturer

A cigarette manufacturer claims that the average nicotine content μ of brand B cigarettes is at most 1.5 mg. It would be unwise to reject the manufacturer's claim without strong contradictory evidence. Suppose the nicotine content is known to be normal with $\sigma=0.2$. Assume $n=10$.

Formulate the hypothesis testing.

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Formulate the hypothesis testing.

$$H_0: \mu = \mu_0 = 1.5$$

$$H_a: \mu > \mu_0 = 1.5$$

Class exercise:

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A cigarette manufacturer claims that the average nicotine content μ of brand B cigarettes is at most 1.5 mg. It would be unwise to reject the manufacturer's claim without strong contradictory evidence. Suppose the nicotine content is known to be normal with $\sigma=0.2$. Assume $n=10$.

- Formulate the hypothesis testing.
- Do you reject H_0 if $\bar{x} = 1.596$? Why or why not?

Tests about a population mean of a normal r.v. with known variance



$$H_0: \mu = 1.5$$

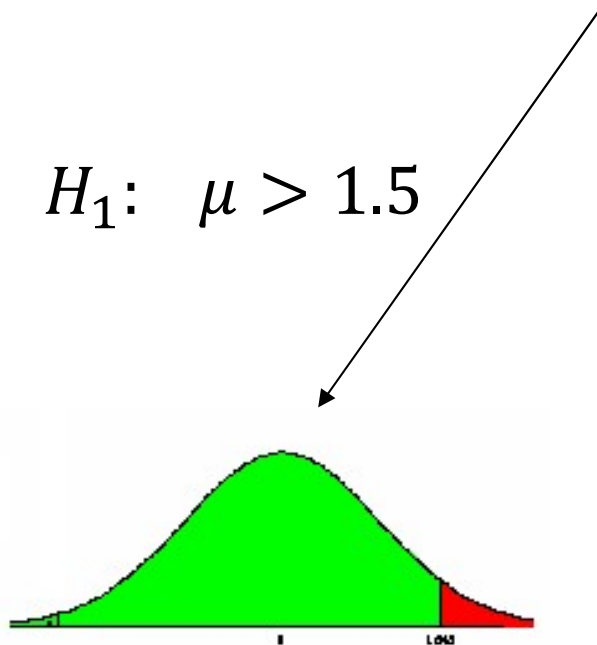
$$\text{Test statistics: } Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

Test statistics is given under the assumption that H_0 is true.

$$H_1: \mu > 1.5$$

 Accept

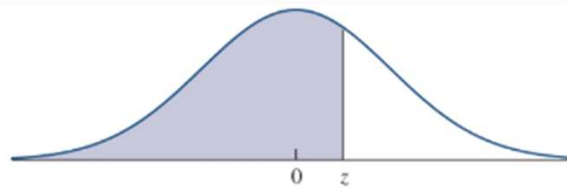
 Reject



$$Z = \frac{1.596 - 1.5}{0.2/\sqrt{10}} = 1.52$$

$$z_{0.05} = 1.645; z_{0.10} = 1.28$$

$$\text{R.R. : } z \geq z_{\alpha}$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986



Class exercise:

Cigarette manufacturer

A cigarette manufacturer claims that the average nicotine content μ of brand B cigarettes is at most 1.5 mg. It would be unwise to reject the manufacturer's claim without strong contradictory evidence. Suppose the nicotine content is known to be normal with $\sigma=0.2$. Assume $n=10$.

- Formulate the hypothesis testing.
- Do you reject H_0 if $\bar{x} = 1.596$? Why or why not?
- If you cannot reject H_0 , what is the probability of a type-II error if $\mu = 1.6$, i.e., $\beta(1.6)$?

We need to know the rejection region for $\bar{X}$ to compute probability of type-2 error.

Example: Cigarette manufacturer

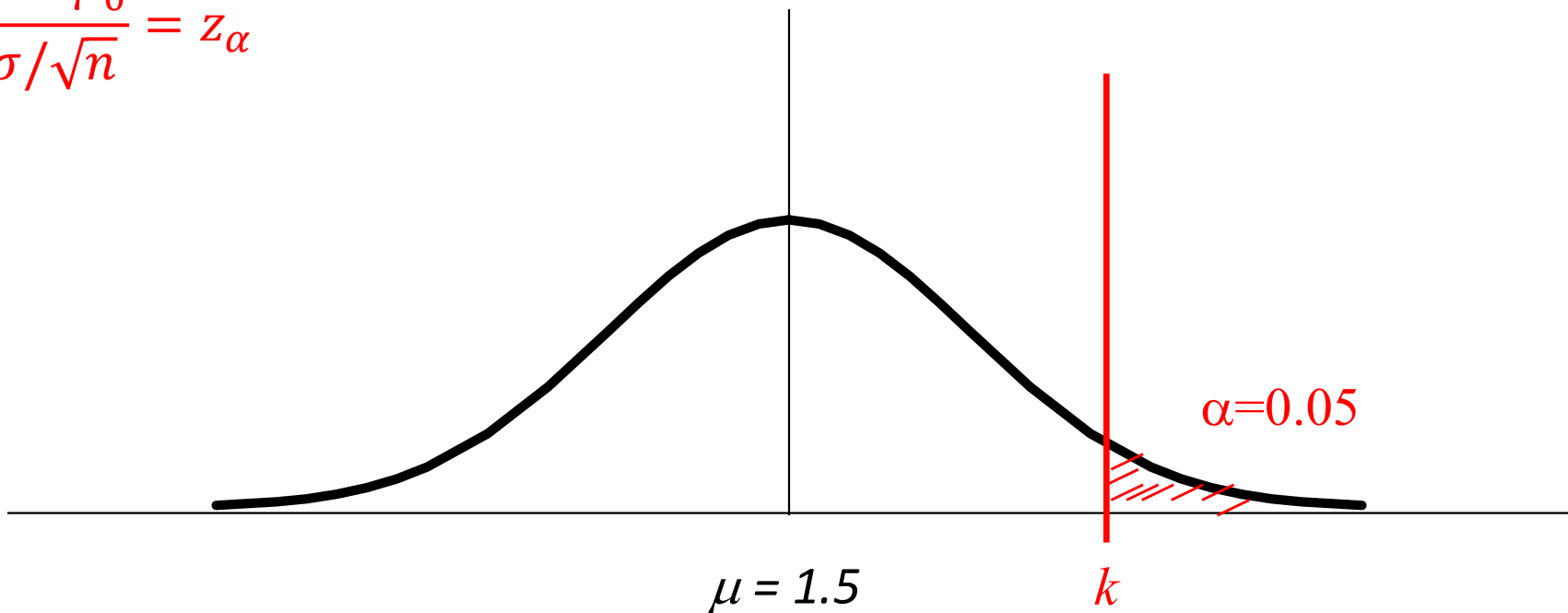
$$H_0: \mu = \mu_0 = 1.5$$

$$H_a: \mu > \mu_0 = 1.5$$

$$\alpha = P\{\text{Type I}\} = P\{\text{reject } H_0 \text{ when } H_0 \text{ is true}\} = P\{\bar{X} > k, \mu = 1.5\}$$

$$\frac{k - \mu_0}{\sigma/\sqrt{n}} = z_\alpha$$

$$\bar{x} = 1.596$$



Class exercise: Cigarette manufacturer

$$H_0: \mu = 1.5$$

$$H_a: \mu > 1.5$$

$$n=10, \alpha=0.05, \sigma=0.2.$$

$$\begin{aligned}\alpha &= P\{\text{type I error}\} = P\{\text{reject } H_0, H_0 \text{ is true}\} \\ &= P\{\bar{X} > k, \mu = \mu_0\} = P\left\{\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} > \frac{k - \mu_0}{\sigma/\sqrt{n}}\right\}\end{aligned}$$

z_α

$$k = \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}} = 1.5 + 1.645 \frac{0.2}{\sqrt{10}} = 1.604$$

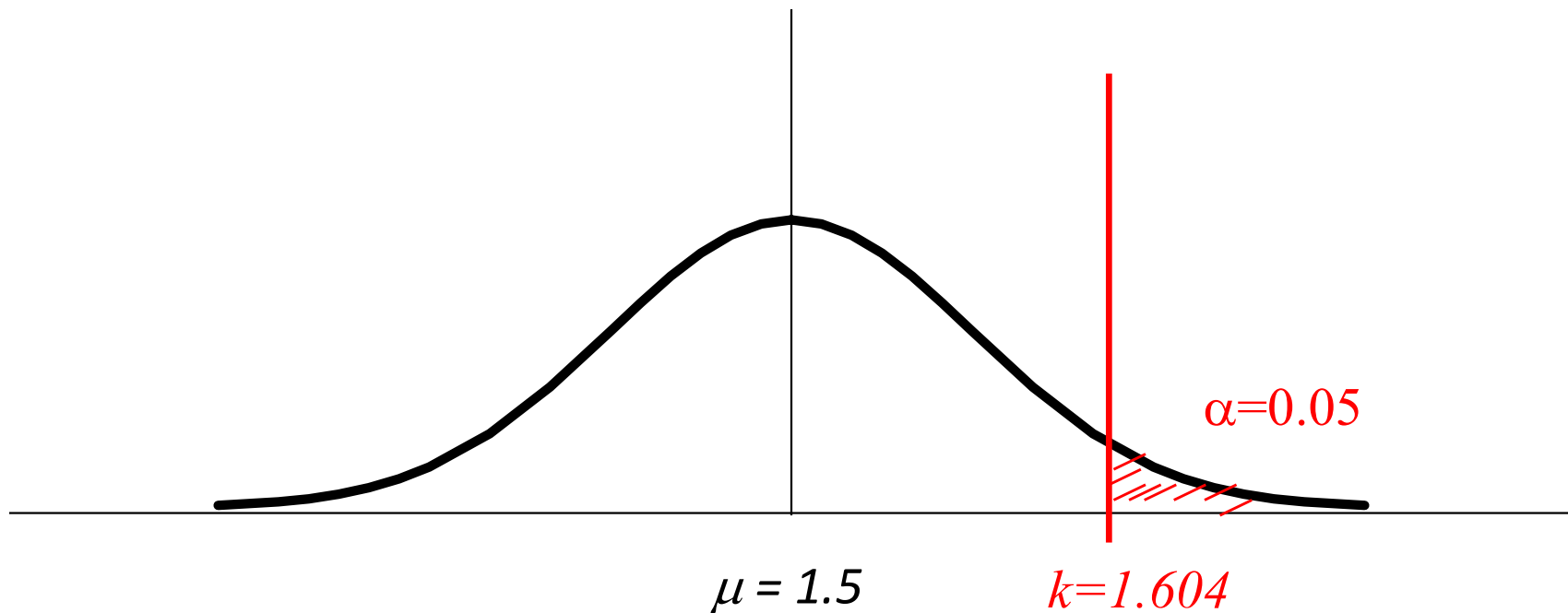
Example: Cigarette manufacturer

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$$\alpha = P\{\text{Type I}\} = P\{\text{reject } H_0 \text{ when } H_0 \text{ is true}\} = P\{\bar{X} > k, \mu = 1.5\}$$

$$\bar{x} = 1.596$$

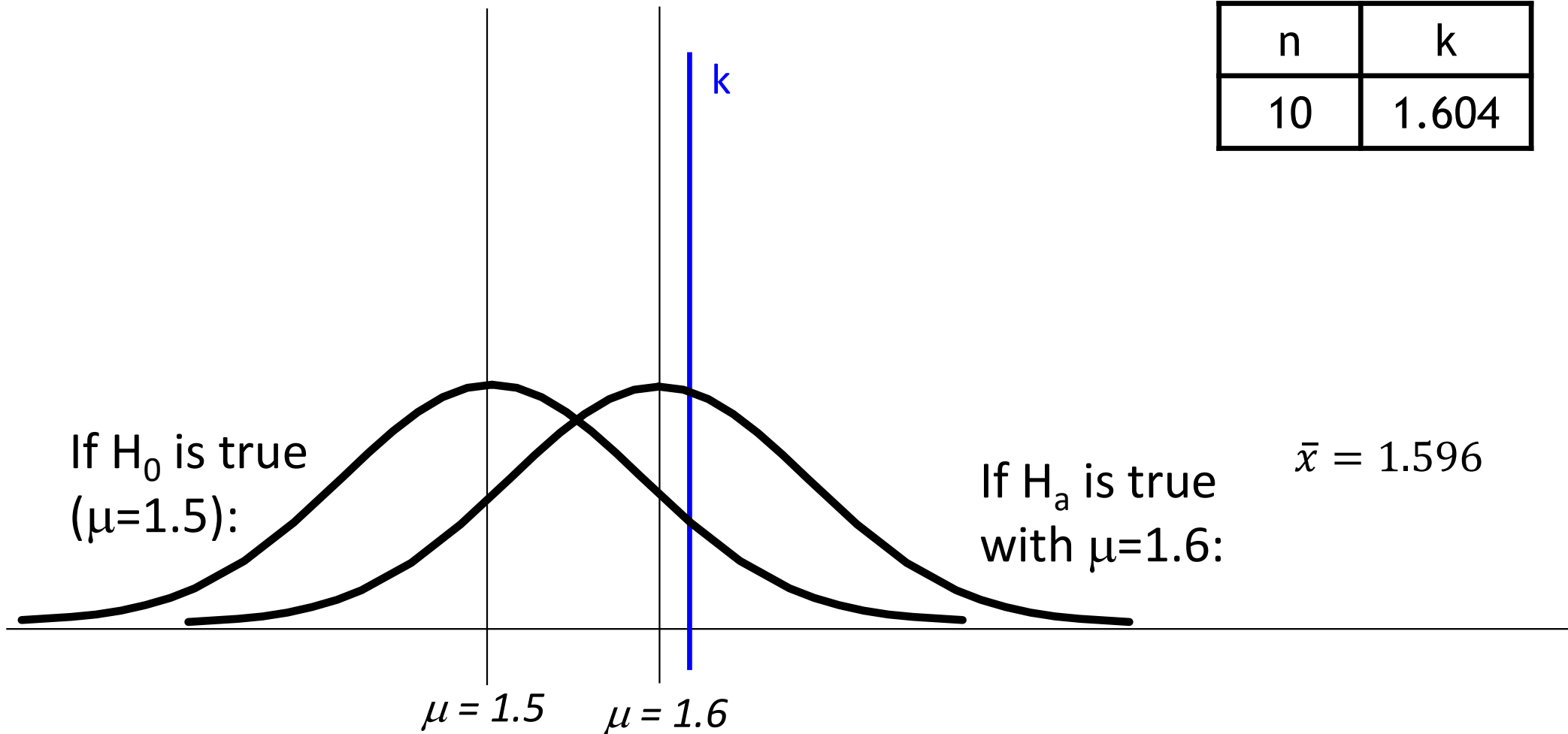


Example: Cigarette manufacturer

$$H_0: \mu = \mu_0 = 1.5$$

$$H_a: \mu > \mu_0 = 1.5$$

n	k
10	1.604

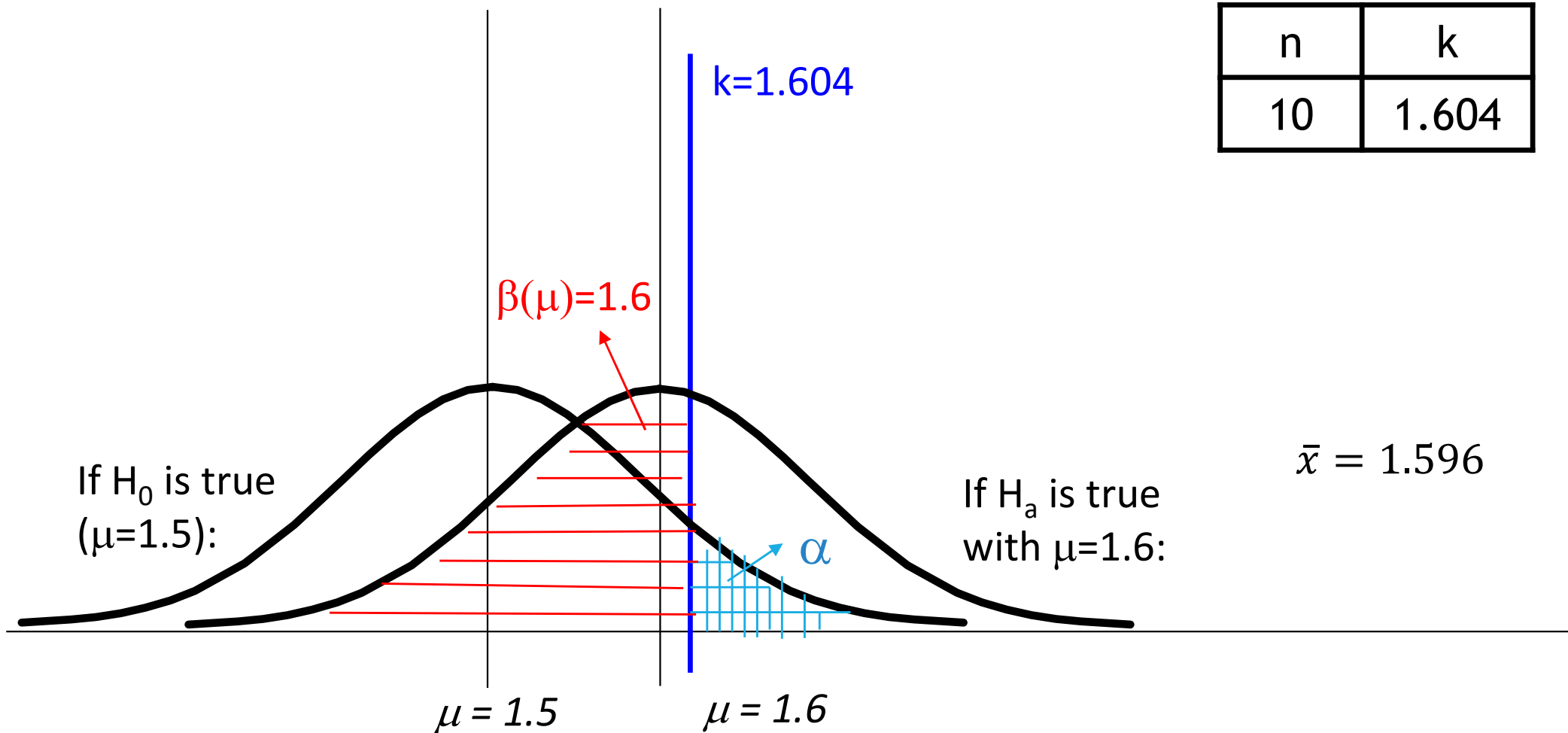


Example: Cigarette manufacturer

$$H_0: \mu = \mu_0 = 1.5$$

$$H_a: \mu > \mu_0 = 1.5$$

n	k
10	1.604



Example: Cigarette manufacturer β and sample size



$$k = \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}}$$

$$H_0: \mu = 1.5$$

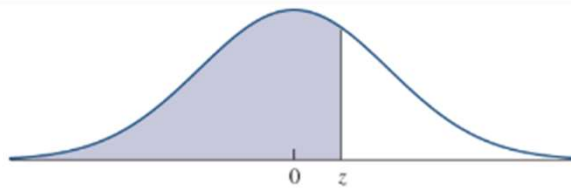
$$H_a: \mu > 1.5$$

$n=10, \alpha=0.05, \sigma=0.2$. What is $\beta(1.6)$? $\beta(1.6) = 0.525$

$$\beta(1.6) = P\{\text{cannot reject } H_0, \mu = 1.6\}$$

$$= P\{\bar{X} < k, \mu = \mu_1\} = P\left\{\frac{\bar{X} - \mu_1}{\sigma/\sqrt{n}} < \frac{\mu_0 + z_\alpha \sigma/\sqrt{n} - \mu_1}{\sigma/\sqrt{n}}\right\}$$

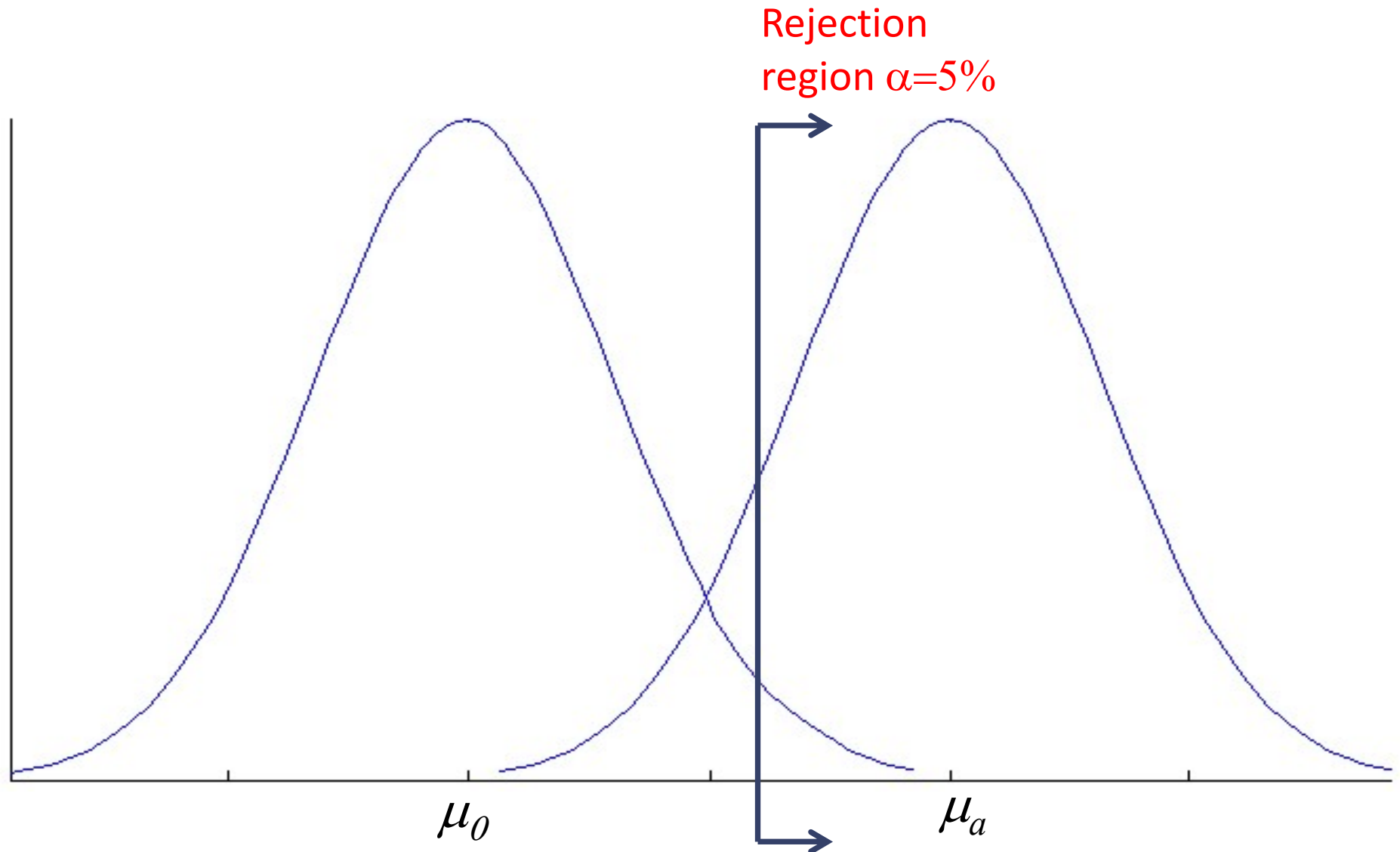
$$= P\left\{Z < z_\alpha + \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}}\right\} \quad -z_\beta = 0.064$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
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2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986



Illustration of rejection region, α and β



Proposition

Suppose an experiment and a sample size are fixed and a test statistic is chosen. Then decreasing the size of the rejection region to obtain a smaller value of α results in a larger value of β for any particular parameter value consistent with H_a .

A way to decrease the probability of both errors?

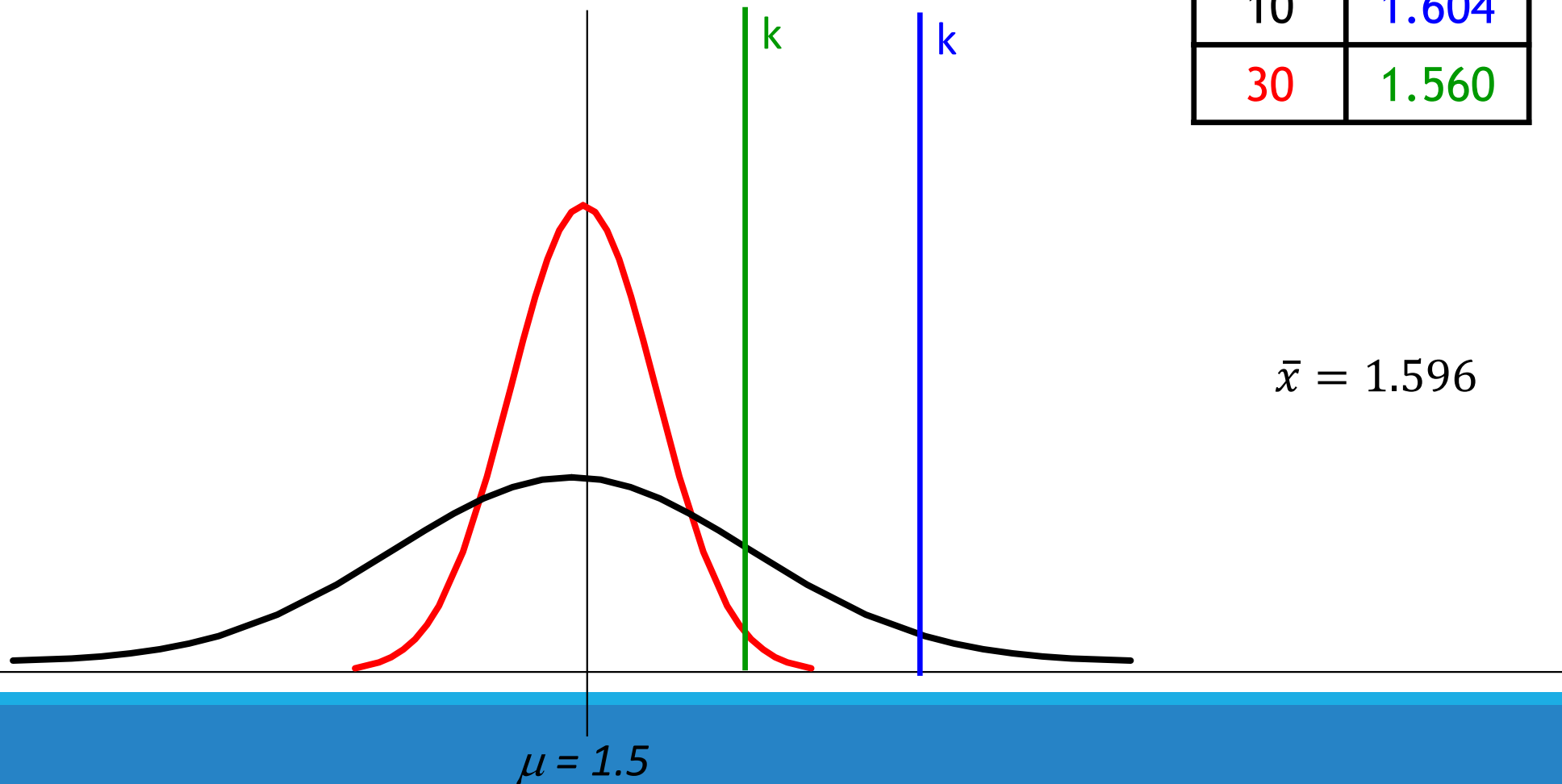
Effect of sample size: Cigarette manufacturer

$$H_0: \mu = \mu_0 = 1.5$$

$$H_a: \mu > \mu_0 = 1.5$$

If H_0 is true and $\alpha=0.05$, $\sigma=0.2$:

n	k
10	1.604
30	1.560



Effect of sample size: Cigarette manufacturer

$$H_0: \mu = \mu_0 = 1.5$$

$$H_a: \mu > \mu_0 = 1.5$$

$$\bar{x} = 1.596$$

$$\beta(1.6) = 0.525$$

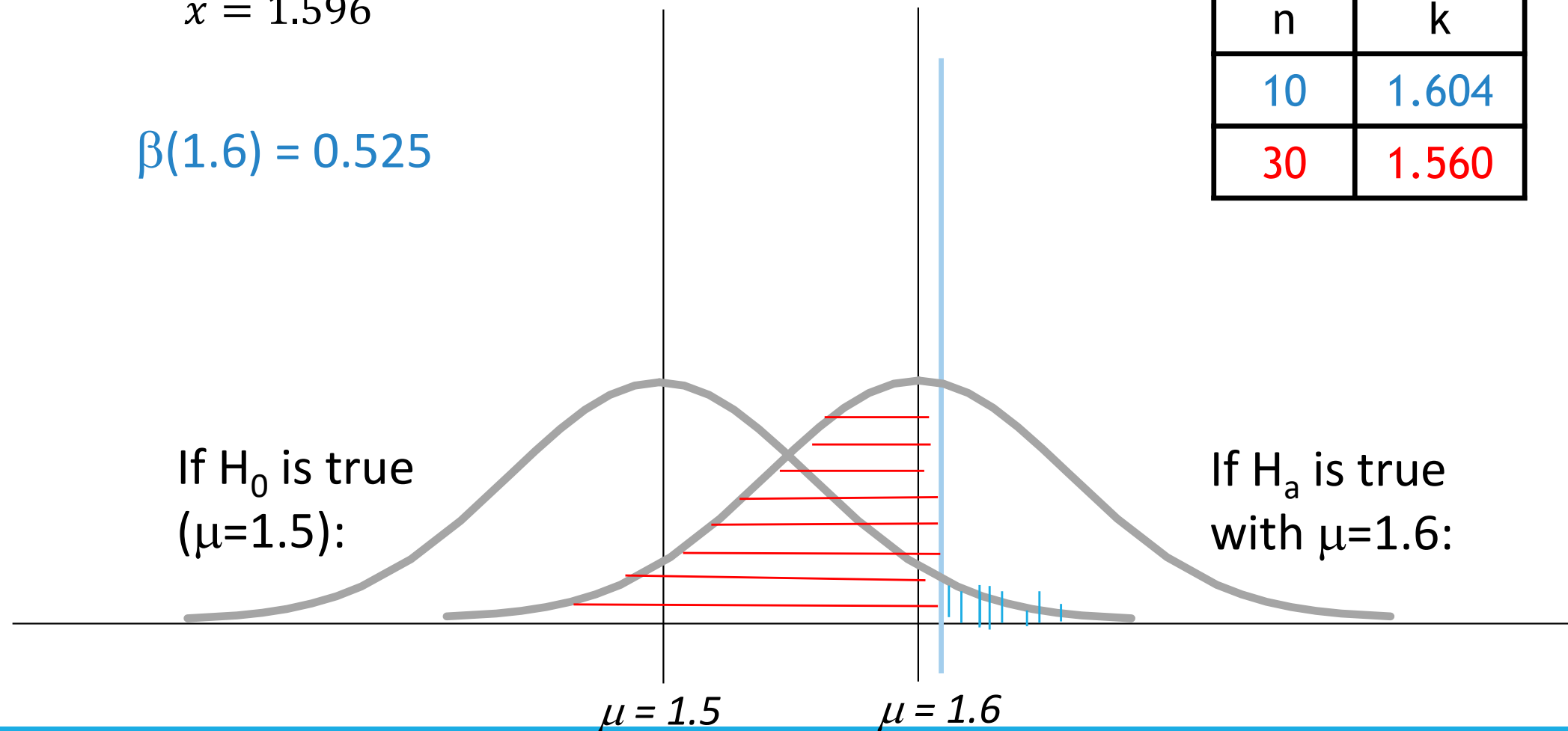
n	k
10	1.604
30	1.560

If H_0 is true
($\mu=1.5$):

If H_a is true
with $\mu=1.6$:

$$\mu = 1.5$$

$$\mu = 1.6$$



Effect of sample size: Cigarette manufacturer

$$H_0: \mu = \mu_0 = 1.5$$

$$H_a: \mu > \mu_0 = 1.5$$

$$\bar{x} = 1.596$$

$$-z_\beta = -1.09$$

$$\Phi(1.09) = 0.8621$$

$$\beta(1.6) = 0.138$$

n	k
10	1.604
30	1.560

If H_0 is true
($\mu=1.5$):

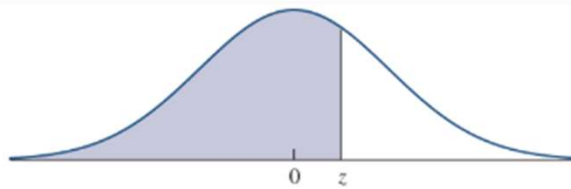
If H_a is true
with $\mu=1.6$:

$$\mu = 1.5$$

$$\mu = 1.6$$



$$z_\beta = z_\alpha + \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}}$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
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2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986



Another approach to hypothesis testing: P -values

Hypothesis testing:

$\alpha \rightarrow \underbrace{\text{Rejection Region}}_{\text{computation of the critical value, } k} \rightarrow \underbrace{\text{conclusion}}_{\text{comparison of } (\bar{x} \text{ and } k) \text{ or } (z \text{ and } z_\alpha)}$

Significance testing:

$\bar{x} \rightarrow \underbrace{P\{\text{type I error when } H_0 \text{ is rejected with } k = \bar{x}\}}_{\text{computation of the } P\text{-value, } P} \rightarrow \underbrace{\text{conclusion}}_{\text{comparison of } P\text{-value and } \alpha}$

Why? Preference on α may change, P -value gives more information.

P-values

- Definition: The p-value (or observed significance level) is the smallest level of significance at which H_0 would be rejected when a specified test procedure is used on a given data set. Once the p-value has been determined, the conclusion at any particular level α results from comparing the P-value to α :
 - 1) P-value $\leq \alpha \rightarrow$ reject H_0 at level α
 - 2) P-value $> \alpha \rightarrow$ cannot reject H_0 at level α

Tests about a population mean of a normal r.v. with known variance (Hypothesis Testing)

$$H_0: \mu = \mu_0$$

$$\text{Test statistics: } Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

Test statistics is given
under the assumption
that H_0 is true.

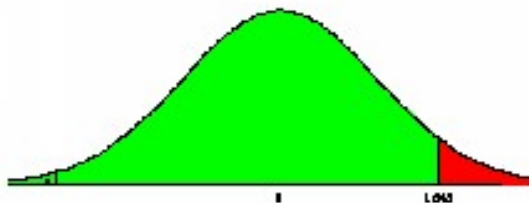
$$H_1: \mu > \mu_0$$

$$H_1: \mu < \mu_0$$

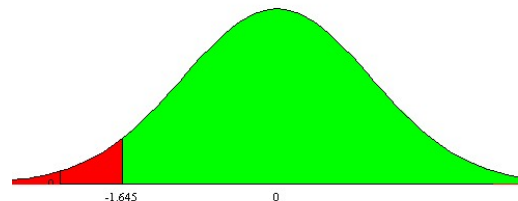
$$H_1: \mu \neq \mu_0$$

Accept

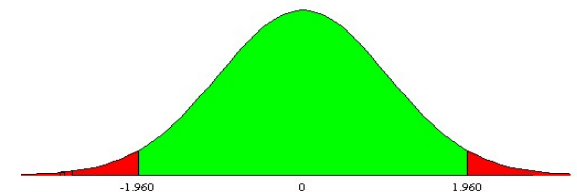
Reject



$$\text{R.R. : } z \geq z_\alpha$$



$$\text{R.R. : } z \leq -z_\alpha$$



$$\begin{aligned} \text{R.R. : } z &\leq -z_{\alpha/2} \\ \text{or } z &\geq z_{\alpha/2} \end{aligned}$$

Tests about a population mean of a normal r.v. with known variance (Significance Testing)

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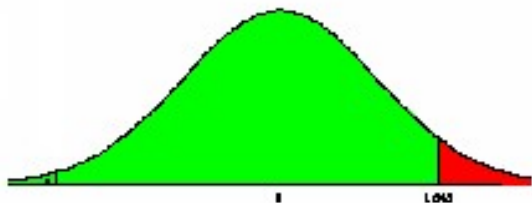
$$H_1: \mu > \mu_0$$

$$H_1: \mu \neq \mu_0$$

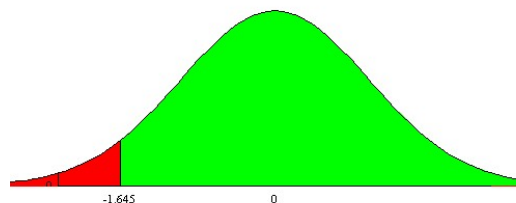
$$H_1: \mu < \mu_0$$

Accept

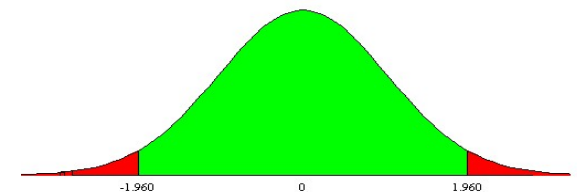
Reject



P-value: $P(Z \geq z_0)$



P-value: $P(Z \leq z_0)$



P-value: $2 \times P(Z \geq |z_0|)$

Tests about a population mean of a r.v. with large sample size

$$H_0: \mu = \mu_0$$

$$\text{Test statistics: } Z = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

Test statistics is given under the assumption that H_0 is true.

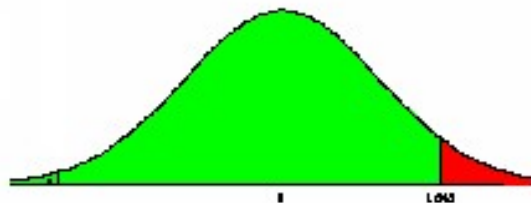
$$H_1: \mu > \mu_0$$

$$H_1: \mu < \mu_0$$

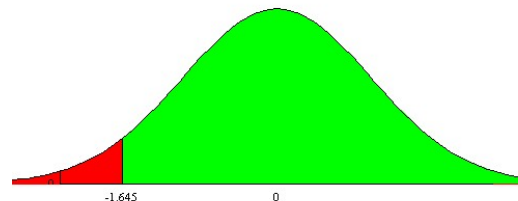
$$H_1: \mu \neq \mu_0$$

Accept

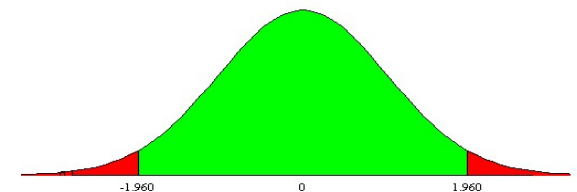
Reject



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$$\begin{aligned} \text{R.R. : } z &\leq -z_{\alpha/2} \\ \text{or } z &\geq z_{\alpha/2} \end{aligned}$$

Tests about a population mean of a r.v. with large sample size (Significance Testing)

$$H_0: \mu = \mu_0$$

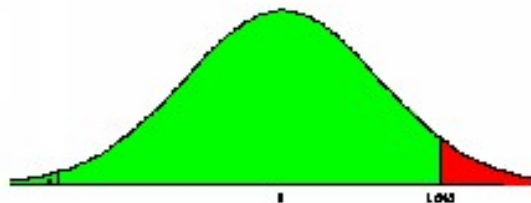
$$\text{Test statistics: } z_0 = \frac{\bar{x} - \mu_0}{S/\sqrt{n}}$$

Test statistics is given under the assumption that H_0 is true.

$$H_1: \mu > \mu_0$$

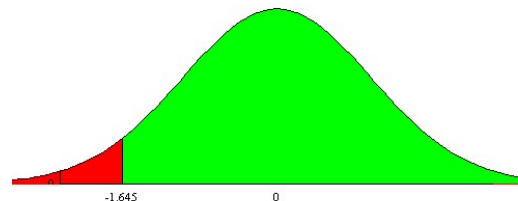
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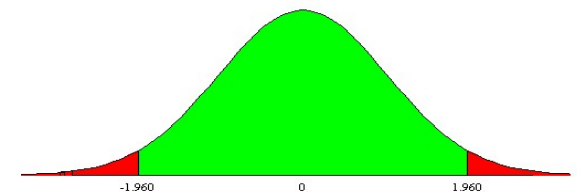
P-value: $P(Z \geq z_0)$

$$H_1: \mu < \mu_0$$



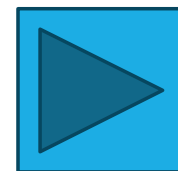
P-value: $P(Z \leq z_0)$

$$H_1: \mu \neq \mu_0$$



P-value: $2 \times P(Z \geq |z_0|)$

Class exercise: Cigarette manufacturer



A cigarette manufacturer claims that the average nicotine content μ of brand B cigarettes is at most 1.5 mg. It would be unwise to reject the manufacturer's claim without strong contradictory evidence. Suppose the nicotine content is known to be normal with $\sigma=0.2$. Assume $n=10$.

$$H_0: \mu = \mu_0 = 1.5$$

$$H_a: \mu > \mu_0 = 1.5$$

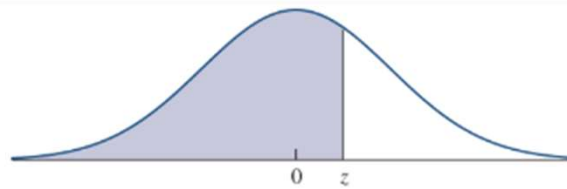
What is the corresponding p-value if $\bar{x} = 1.596$?

$$Z = \frac{1.596 - 1.5}{0.2/\sqrt{10}} = 1.52$$



$$\begin{aligned} \text{P-value} &= 1 - P(1.52) = \\ &= 1 - 0.9357 = 6.43\% \end{aligned}$$

Corresponding decisions?



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
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2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
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2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986



Class exercise

The target thickness for silicon wafers used in a specific type of integrated circuit is 245 mm. A sample of 50 wafers is obtained, and the thickness of each is determined, yielding a sample mean of 246.18 mm and a sample standard deviation of 3.6 mm. Does this data suggest that the true average wafer thickness is something other than the target value?

Formulate the hypothesis test.

Compute the p-value.

What is your conclusion when $\alpha=1\%$ and $\alpha=5\%$?

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Formulate the hypothesis test.

$$H_0: \mu = \mu_0 = 245$$

$$H_a: \mu \neq \mu_0 = 245$$

Class exercise

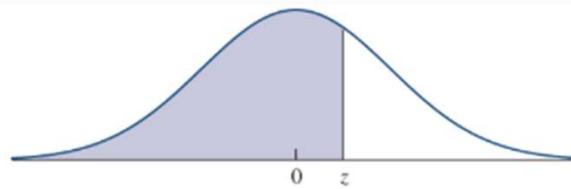
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Compute the p-value.

$$Z = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} = \frac{246.18 - 245}{3.6/\sqrt{50}} = 2.32$$

$$\Phi(2.32) = 0.9898$$

$$\text{P-value} = 2 * (1 - \Phi(2.32)) = 2.04\%$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
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$$\text{P-value} = 2 * (1 - \Phi(2.32)) = 2.04\%$$

Hypothesis testing versus CIs

$$H_0: \mu = \mu_0$$

$$H_a: \mu \neq \mu_0$$

at significance level α

If μ_0 is in (l,u) , do we reject H_0 ? Why?

2-sided confidence

interval (l,u) at

$100(1-\alpha)\%$